EXERCISE 5a

Simulating the Lifecycle Stages for UI Design Using the RAD Model

and Development of a Small Interactive Interface Using Axure RP

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AIM

The aim of this exercise is to simulate the lifecycle stages of UI design using the Rapid Application Development (RAD) model, and to develop a small interactive interface prototype employing Axure RP as the prototyping tool.

TOOLS REQUIRED

- Axure RP (Download from: https://www.axure.com/)
- Web browser (for prototype preview)
- Axure Cloud (optional, for sharing prototypes)

THEORY

1. RAD Model – Rapid Application Development

The Rapid Application Development (RAD) model is an adaptive software development methodology that prioritizes speed, iterative development, and continuous user feedback over rigid sequential processes. Unlike traditional waterfall approaches, RAD allows teams to build, test, and refine prototypes in short cycles, making it especially suited for UI and front-end design projects where user feedback plays a critical role in shaping the final product.

The RAD model consists of four primary phases:

Phase 1: Requirements Planning

In this phase, the project team collaborates with stakeholders to identify the system's goals, constraints, and high-level functional requirements. Unlike detailed specification documents used in waterfall models, RAD requires only a broad understanding of what needs to be built. Key activities include:

- Identifying the core features of the UI
- Engaging stakeholders to capture needs and expectations
- Defining scope and prioritizing features for early iterations

Phase 2: User Design

This is the most iterative phase of the RAD model. Designers create initial wireframes and interactive prototypes based on the gathered requirements, present them to users, collect feedback, and refine the designs accordingly. Tools such as Axure RP are heavily used in this phase. Key activities include:

- Developing wireframes for each screen of the application
- Adding interactivity through click events and navigation flows
- Conducting usability tests and incorporating user feedback
- Creating reusable masters for common UI components (header, footer)

Phase 3: Construction

Based on the approved prototype, the actual functional UI is developed. Developers convert high-fidelity mockups into working code. Iterative build-test-feedback cycles continue during this phase to ensure alignment with user expectations. Key activities include:

- Converting Axure prototypes into working HTML/CSS/JavaScript
- Integrating dynamic panels, pop-ups, and carousel interactions
- Performing rounds of user testing and fixing defects

Phase 4: Cutover

In this final phase, the completed application is deployed to the production environment. End users are trained on the new system and ongoing support is provided to address any post-launch issues. Key activities include:

- Deploying the final UI to the live environment
- Conducting user training sessions
- Providing technical documentation and post-launch support

PROCEDURE

The following steps describe the complete procedure for developing a Shopping App interface using Axure RP, following the RAD model lifecycle.

Phase 1: Requirements Planning

Step 1 – Identify Key Features

Navigation screens to be designed:

- Home Page
- Product Categories
- Product Listings
- Product Details

- Shopping Cart
- Checkout
- Order Confirmation
- Order History

User actions to be supported:

- Browsing and searching for products
- Adding items to the cart
- Completing the checkout process
- Tracking orders via the Order History screen

Step 2 – Create a Requirements Document

- List all features and functionalities agreed upon with stakeholders.
- Document user stories (e.g., "As a user, I want to search for products by category").
- Define use cases for each primary user action.

Phase 2: User Design (Axure RP Workflow)

Step 1 – Install and Launch Axure RP

1. Visit https://www.axure.com/ and download the latest version of Axure RP.
2. Run the installer and follow the on-screen instructions.
3. Launch Axure RP after installation.

Step 2 – Create a New Project

4. Go to File → New to create a new project.
5. Name the project "Shopping App Interface".
6. Set the canvas size appropriate for web or mobile layout (e.g., 1280×800 for web).

Step 3 – Create Wireframes

7. Use the widget library on the left panel to drag and drop elements (buttons, text boxes, images, containers) onto the canvas.
8. Design a separate wireframe page for each screen: Home Page, Product Categories, Product Listings, Product Details, Cart, Checkout, Order Confirmation, and Order History.
9. Maintain a consistent layout structure across all screens for a cohesive user experience.

Step 4 – Add Interactions

10. Select a widget (e.g., the "Add to Cart" button) and open the Properties panel on the right.

11. Click on Interactions and choose an event (e.g., OnClick).
12. Define the action for the event (e.g., navigate to the Cart screen).
13. Repeat for all interactive elements across all screens.

Step 5 – Create Masters (Reusable Components)

14. In the Masters panel, create reusable components such as the Header, Footer, and Navigation Bar.
15. Design these components once and drag them onto all relevant wireframe pages.
16. Any updates made to a master will automatically reflect across all pages using it.

Step 6 – Add Annotations

17. Select each widget and use the Notes panel to add a description of its purpose and expected behavior.
18. Annotations are useful for communicating design intent to developers and testers.

Phase 3: Construction (Interactive Prototyping)

Step 1 – Develop Interactive Prototypes

19. Enhance wireframes with transitions and animations (e.g., slide-in, fade-in) for navigation between screens.
20. Use Dynamic Panels to build interactive components such as image carousels on the Home Page, pop-up dialogs for add-to-cart confirmations, and expandable product descriptions.
21. Set up conditional logic where needed (e.g., show error message if cart is empty during checkout).

Step 2 – Test and Iterate

22. Click the Preview button to run the prototype in a web browser.
23. Walk through all user flows and test every interactive element.
24. Note usability issues or broken interactions and refine accordingly.
25. Share the prototype with peers or stakeholders and collect structured feedback.

Phase 4: Cutover (Finalize and Export)

Step 1 – Finalize and Export

26. After all iterations are complete and stakeholders have approved the design, finalize all interactions and visual styles.
27. Export the prototype as an HTML file: Go to Publish → Generate HTML Files.
28. Alternatively, publish directly to Axure Cloud for easy sharing via a URL.

Step 2 – User Training and Support

29. Conduct a walkthrough session with end users to familiarize them with the new interface.
30. Provide supporting documentation including a user guide and annotated screenshots.
31. Offer a support channel for post-deployment issues and change requests.

RESULT

The lifecycle stages of UI design were successfully simulated using the Rapid Application Development (RAD) model. A small interactive Shopping App interface was designed and prototyped using Axure RP, incorporating screens for Home, Product Categories, Product Listings, Product Details, Cart, Checkout, Order Confirmation, and Order History. Interactive navigation and reusable master components were implemented and tested through iterative feedback cycles.